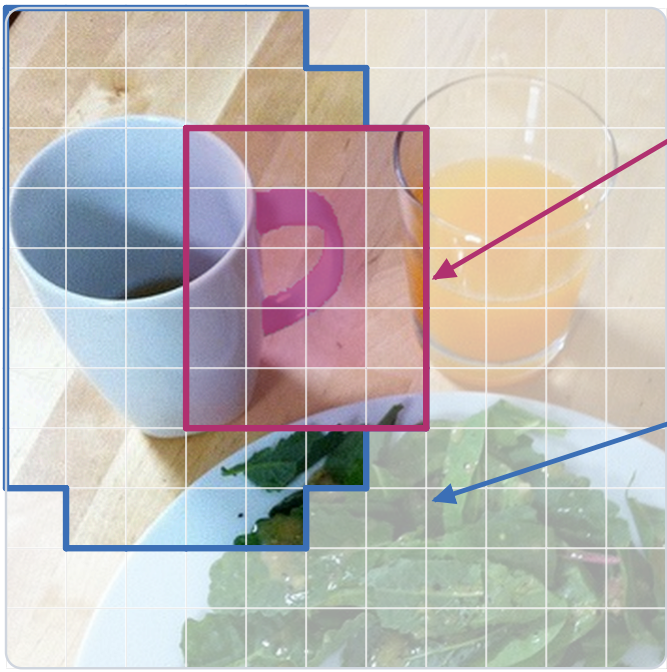
(a) Region of a mask token on the image-token grid



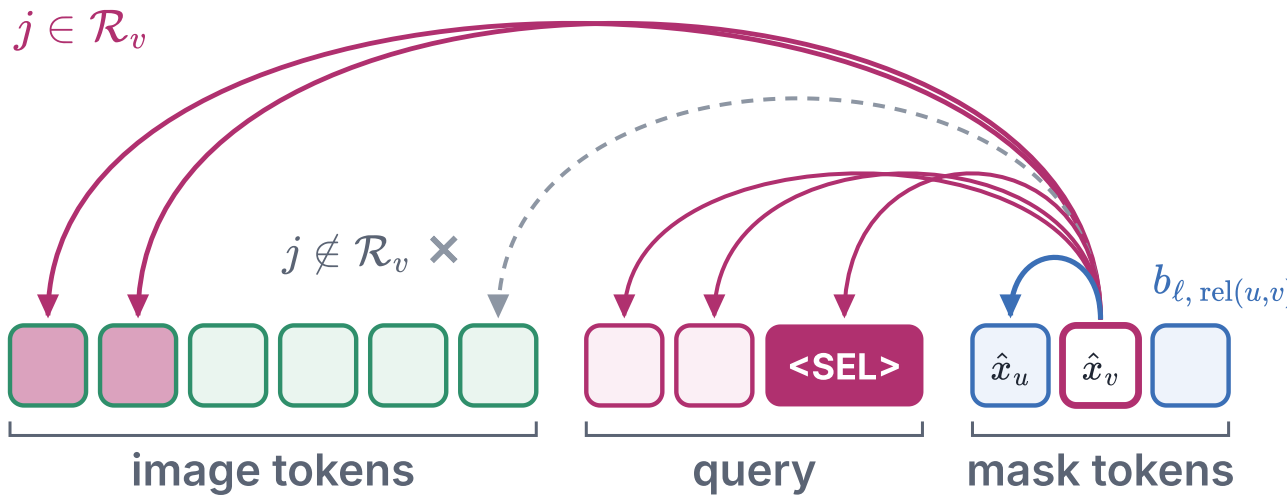
$\hat{x}_v$ sees $\mathcal{R}_v$

cells under m_v
+ 1-cell margin

$\hat{x}_u$ sees $\mathcal{R}_u$

region of the parent
mask (the mug)

(b) Attention of mask token v



Attention mask of mask token v

- $e_j,\ j \in \mathcal{R}_v$: 0 $\quad(\mathcal{R}_v =$ cells under m_v, dilated by one cell$)$
- query tokens, $\langle$SEL$\rangle$: 0
- $\hat{x}_u$: $b_{\ell,\,\mathrm{rel}(u,v)}$ (learned, by tree relation)
- $e_j,\ j \notin \mathcal{R}_v$: $-\infty$
- image and query tokens $\to \hat{x}_v$: $-\infty$